

# Fundamentals of Electric Circuits

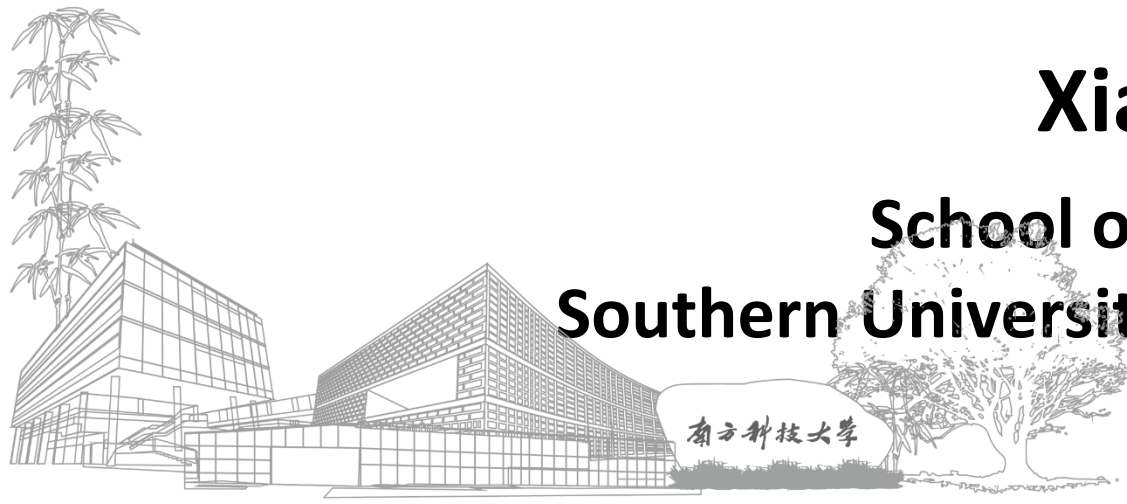
## 电路基础

### Lecture5: Operational Amplifiers

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# Outline

- **Operational Amplifiers (Op Amps)**
- **Ideal Op Amp**
- **Inverting Amplifier**
- **Noninverting Amplifier**
- **Summing Amplifier**
- **Difference Amplifier**
- **Cascaded Op Amp Circuits**
- **Applications**

# Operational Amplifiers: Introduction (I)

- Having learned the basic laws and theorems for circuit analysis, we are now ready to study an active circuit element of paramount importance: the **operational amplifier** (运算放大器), or **op amp** for short. The op amp is an electronic unit that behaves like a **voltage-controlled voltage source**
- With proper resistors and/or capacitors, as well as power supply, connected to its terminals, an operational amplifier can perform mathematical operations like **addition, subtraction, multiplication, division, differentiation, and integration**. The ability of the op amp to perform these **mathematical operations** (数学运算) gives rise to the name
- A full discussion of what is inside the op amp is beyond the scope of this course
- It will suffice to treat the op amp as a **circuit building block** (组成模块) and **simply study what takes place at its terminals** (端口)

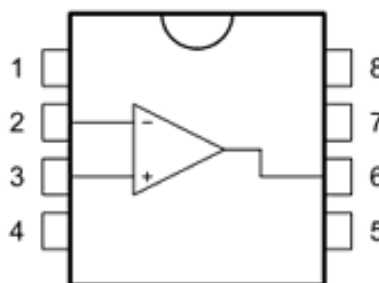
# Operational Amplifiers: Introduction (II)

- Op amps are commercially available in integrated circuit packages in several forms. A typical one is the eight-pin **dual in-line package (or DIP, 双列直插式封装)**
- Pin or terminal 8 is unused, and terminals 1 and 5 are of little concern to us. The five important terminals are: **the inverting input, the noninverting input, the output, the positive and negative power supplies**



Dual In-line Package  
(DIP, 双列直插式封装)

Pin layout of 741

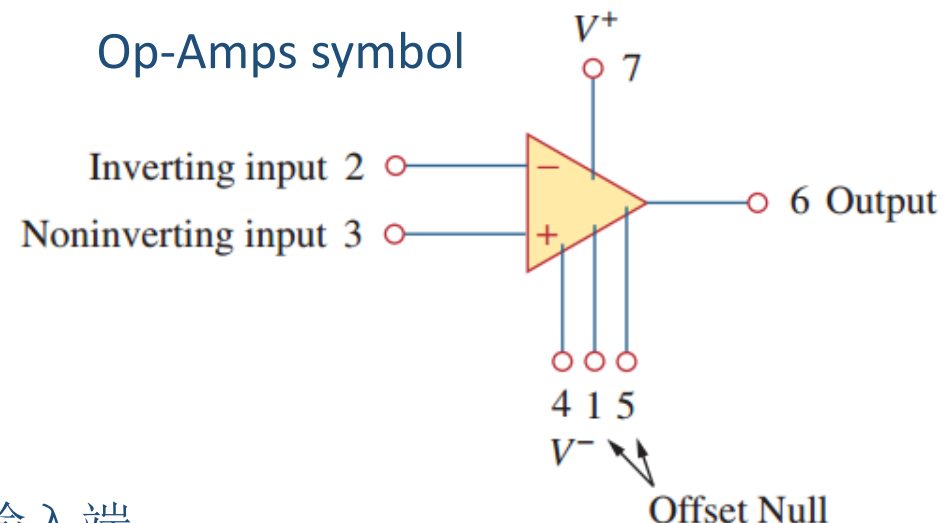


Pin 2: 反相输入端  
Pin 6: 输出端

1 - Offset null 1  
2 - Inverting input  
3 - Non-inverting input  
4 -  $V_{CC}^-$   
5 - Offset null 2  
6 - Output  
7 -  $V_{CC}^+$   
8 - N.C.

Pin 3: 同相输入端  
Pin 7, 4: 正, 负供电电源

Op-Amps symbol

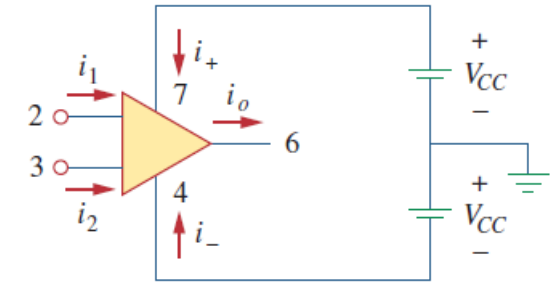


# Basics of Op-Amp (I)

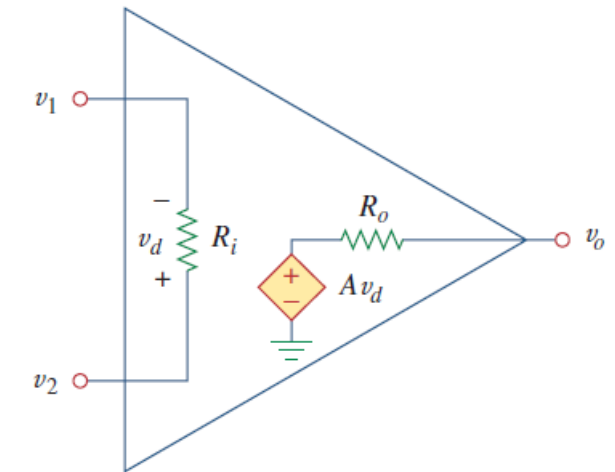
- As an active element, the op amp requires a power source. Most op amps use two voltage sources, with a ground reference between them → This gives a positive and negative supply voltage.
- Often in circuit diagrams the power supply terminals are obscured. It is taken for granted that they must be connected
- The voltage **output** of an op amp is **proportional** to the difference between the noninverting and inverting inputs

$$v_o = Av_d = A(v_2 - v_1) \quad -V_{CC} \leq v_o \leq V_{CC}$$

- A is called the **open-loop gain** (开环增益)
- The output can not exceed the maximum or minimum voltage of the power supply



**Figure 5.3**  
Powering the op amp.



**Figure 5.4**  
The equivalent circuit of the nonideal op amp.

# Basics of Op-Amp (II)

- Ideally, the open-loop gain is infinite. In reality it is high:  $10^5$  to  $10^8$  V/V or 100 to 160 dB
  - Sometimes, voltage gain is expressed in decibels (dB)

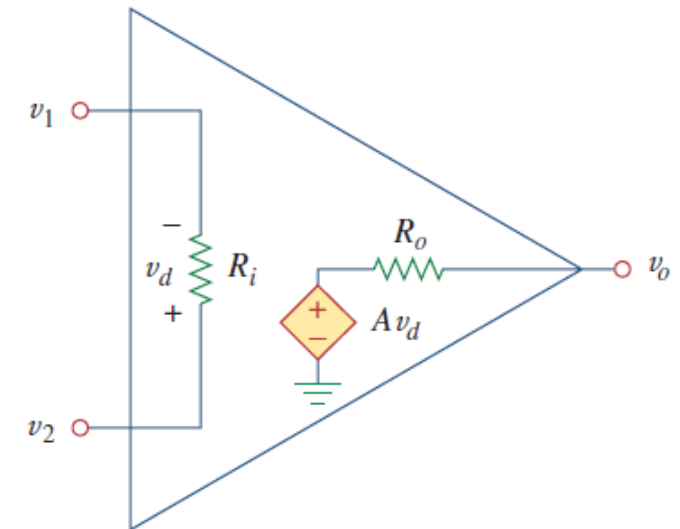
$$A \text{ dB} = 20 \log_{10} A$$

**TABLE 5.1**

Typical ranges for op amp parameters.

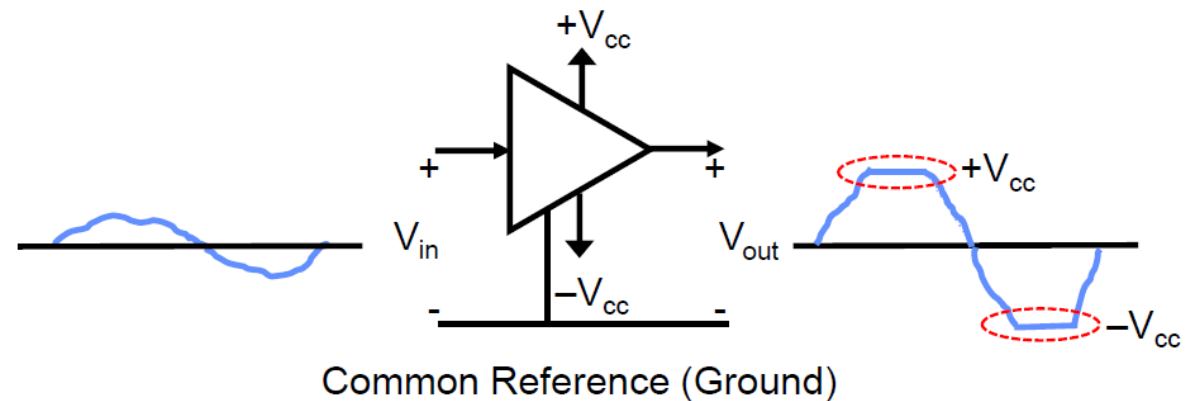
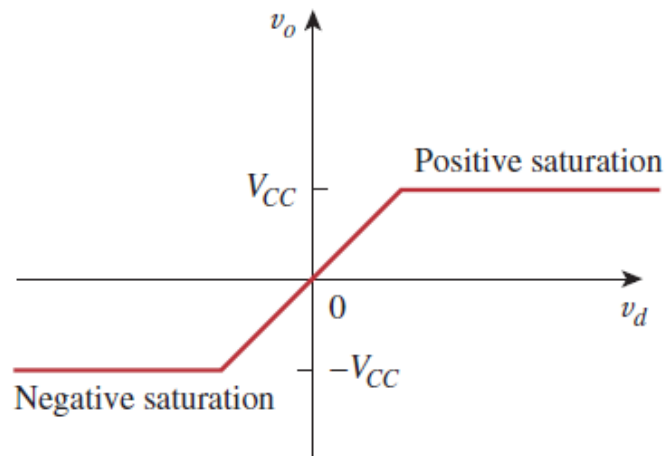
| Parameter                | Typical range              | Ideal values    |
|--------------------------|----------------------------|-----------------|
| Open-loop gain, $A$      | $10^5$ to $10^8$           | $\infty$        |
| Input resistance, $R_i$  | $10^5$ to $10^{13} \Omega$ | $\infty \Omega$ |
| Output resistance, $R_o$ | 10 to $100 \Omega$         | $0 \Omega$      |
| Supply voltage, $V_{CC}$ | 5 to 24 V                  |                 |

开环增益  
输入电阻  
输出电阻  
供电电压



# Operational Amplifiers: Saturation

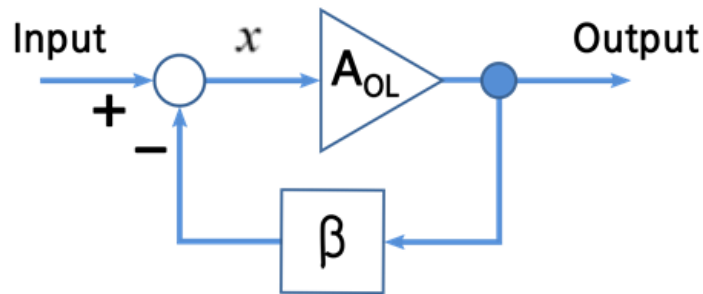
- As an ideal source, the output voltage would be unlimited. In reality, one cannot expect the output to exceed the supply voltages
- When an output should exceed the possible voltage range, the output remains at either the maximum or minimum supply voltage. This is called **saturation (饱和)**
  - If the output wants to go beyond the supplies (for example when the input is positive and large) then it will be clipped at  $+V_{CC}$
- Outputs between these limiting voltages are referred to as the **linear region (线性区)**



*Q: How to operate an amplifier in its linear region?*

# Operational Amplifiers: Negative Feedback

- Op amps take on an expanded functional ability with the use of **feedback (反馈)**
- When there is a feedback path from output to input, the ratio of the output voltage to the input voltage is called the **closed-loop gain (闭环增益)**
- Feedback to the inverting terminal is called “**negative feedback (负反馈)**”
  - The closed-loop gain is almost insensitive to the open-loop gain  $A$  of the op amp
  - Op amps are used in circuits with feedback paths



$$A_{Close\_Loop} = \frac{A_{OL}}{1 + \beta \cdot A_{OL}} \cong \frac{1}{\beta} \text{ when } \beta \cdot A_{OL} \gg 1$$

$$x = Input - \beta \cdot Output; \quad Output = A_{OL} \cdot x$$

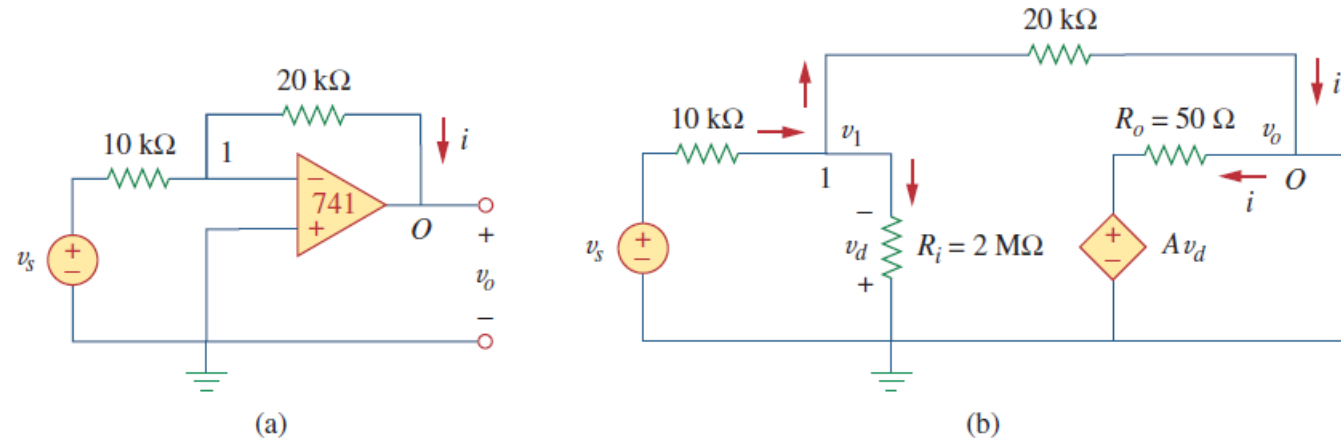
$$Output = \frac{A_{OL}}{1 + \beta \cdot A_{OL}} Input$$

$$x = \frac{1}{1 + \beta \cdot A_{OL}} Input = \frac{Output}{A_{OL}}$$



# Example: Operational Amplifiers

- Q: The op amp 741 in figure below has an open-loop voltage gain of  $2 \times 10^5$ , input resistance of  $2 \text{ M}\Omega$ , and output resistance of  $50 \Omega$ . Find the closed-loop gain  $v_o/v_s$  and the current  $i$  when  $v_s = 2 \text{ V}$



**Figure 5.6**

For Example 5.1: (a) original circuit, (b) the equivalent circuit.

## Solution:

We first build the **equivalent circuit** as shown in Fig. 5.6(b) and then apply **nodal analysis**

At node 1, KCL gives

$$\frac{v_s - v_1}{10 \times 10^3} = \frac{v_1}{2000 \times 10^3} + \frac{v_1 - v_o}{20 \times 10^3}$$

At node O, KCL gives

$$\frac{v_1 - v_o}{20 \times 10^3} = \frac{v_o - Av_d}{50}$$

Since  $v_d = -v_1$ ,  $A = 2 \times 10^5$ , we can solve for the closed-loop gain as  $v_o/v_s = -1.9999699$  ( $\approx -2$ ).

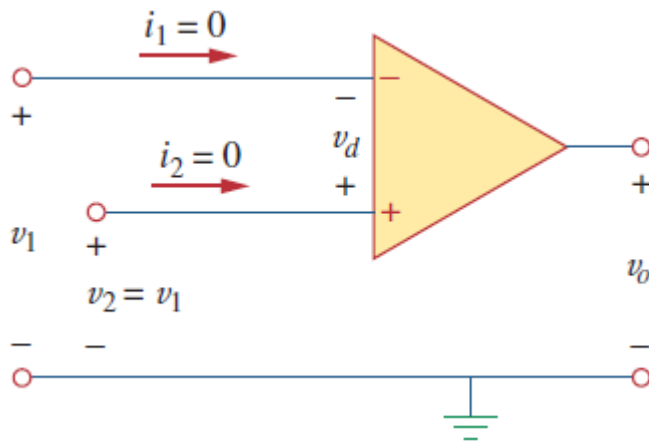
When  $v_s = 2 \text{ V}$ ,  $v_o = -3.9999398 \text{ V}$

$v_1 = 20.066667 \mu\text{V}$

so,  $i = (v_1 - v_o)/20\text{k} = 0.1999 \text{ mA}$

# Ideal Op Amp

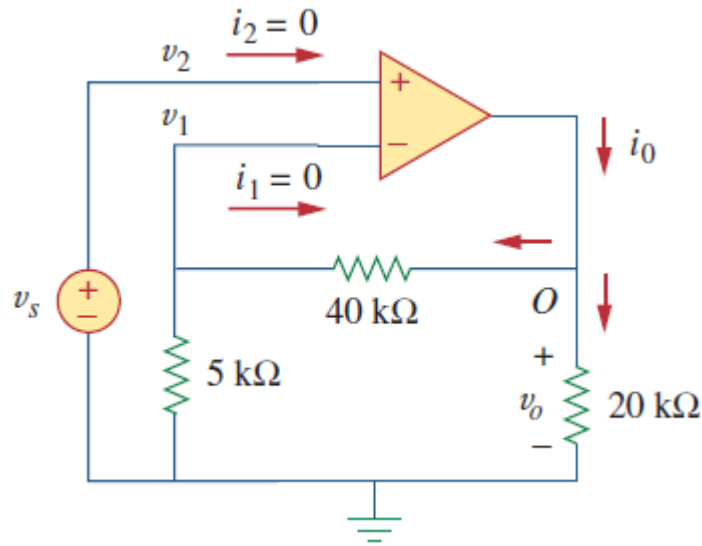
- To facilitate the understanding of op amp circuits, we will assume ideal op amps
  - It has an infinite open-loop gain, i.e.,  $A = \infty$
  - The resistance of the two inputs is infinite, i.e.,  $R_i = \infty$
  - Zero output impedance, i.e.,  $R_o = 0$
- Two extremely important characteristics of the ideal op amp operating in linear mode
  - The currents into both input terminals are zero:  $i_1 = i_2 = 0 \rightarrow$  **Virtual opening (虚断)**
  - The two inputs have the same voltage, i.e.,  $v_1 = v_2 \rightarrow$  **Virtual shorting (虚短)**



- These two features should be regarded as the key handles to analyzing op amp circuits

# Example: Ideal Op Amp

- Q: Using the ideal op amp model, calculate the closed-loop gain  $v_o/v_s$ . Find  $i_o$  when  $v_s = 1$  V



Solution:

The **infinite gain** assumption leads to  $v_1 = v_2 = v_s$

Since according to the **infinite  $R_i$**  assumption,  $i_1 = i_2 = 0$

KCL at node  $v_1$  gives  $v_s/5k + (v_s - v_o)/40k = 0$

Therefore,  $v_o/v_s = 9$

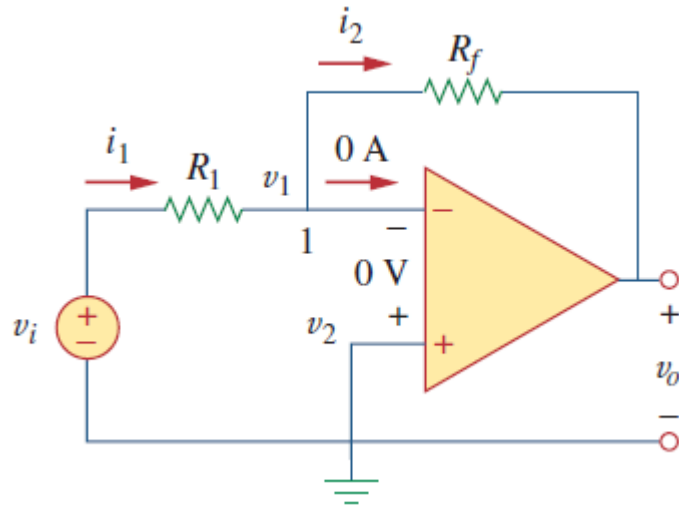
When  $v_s = 1$  V,  $v_o = 9$  V, and hence  $i_o = v_o/20k + (v_o - v_s)/40k = 0.65$  mA

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# Inverting Amplifier

- We will consider some useful op amp circuits that often serve as modules for designing more complex circuits
- The first of such op amp circuits is the **inverting amplifier (反相放大器)**
  - The noninverting input is grounded
  - The inverting terminal is connected to the output via a feedback resistor,  $R_f$
  - The input is also connected to the inverting terminal via another resistor,  $R_1$



- Applying KCL to node 1, yielding

$$i_1 = i_2 \Rightarrow \frac{v_i - v_1}{R_1} = \frac{v_1 - v_o}{R_f}$$

$$\frac{v_i}{R_1} = -\frac{v_o}{R_f}$$

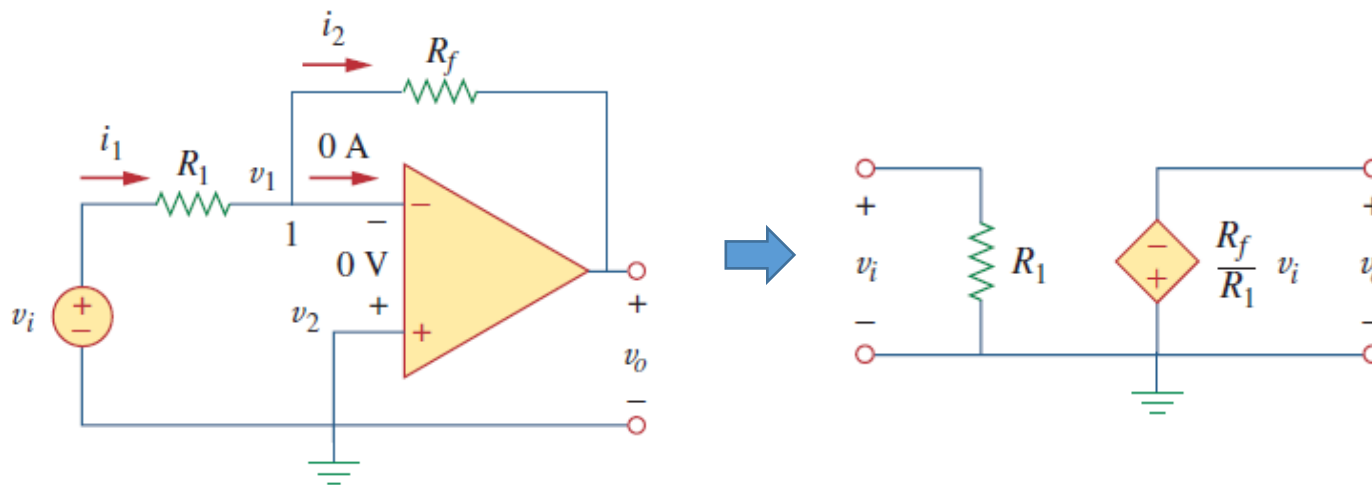


$$v_o = -\frac{R_f}{R_1} v_i$$

*Both the input signal and the feedback are applied at the inverting terminal of the op amp*

# Inverting Amplifier: Equivalent Circuit

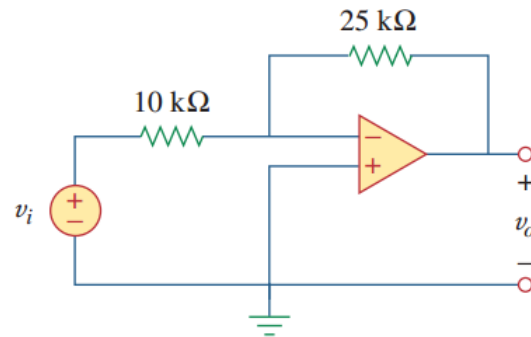
- From this closed-loop gain expression one can see that
  - The gain is the ratio of the feedback resistor  $R_f$  and  $R_1$ , i.e., the gain depends only on the external elements connected to the op amp
  - The polarity of the output is the reverse of the input, thus the name “inverting” amplifier
- The inverting amplifier’s equivalent circuit is shown below. Note that  $R_1$  is the input resistance of the inverting amplifier



- It is a good candidate for making a current-to-voltage converter (电流-电压转换器), i.e., converting the input current into an output voltage for further processing

# Example: Inverting Amplifier

**Example 5.3** Refer to the op amp in Fig. 5.12. If  $v_i = 0.5$  V, calculate: (a) the output voltage  $v_o$ , and (b) the current in the 10-k $\Omega$  resistor.



**Figure 5.12**  
For Example 5.3.

**Solution:**

(a) Using Eq. (5.9),

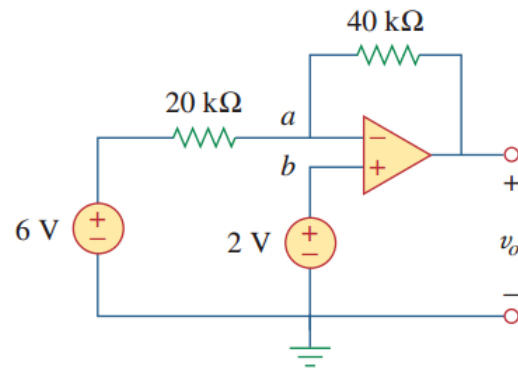
$$\frac{v_o}{v_i} = -\frac{R_f}{R_1} = -\frac{25}{10} = -2.5$$

$$v_o = -2.5v_i = -2.5(0.5) = -1.25 \text{ V}$$

(b) The current through the 10-k $\Omega$  resistor is

$$i = \frac{v_i - 0}{R_1} = \frac{0.5 - 0}{10 \times 10^3} = 50 \mu\text{A}$$

**Example 5.4** Determine  $v_o$  in the op amp circuit shown in Fig. 5.14.



**Figure 5.14**  
For Example 5.4.

**Solution:**

Applying KCL at node  $a$ ,

$$\frac{v_a - v_o}{40 \text{ k}\Omega} = \frac{6 - v_a}{20 \text{ k}\Omega}$$

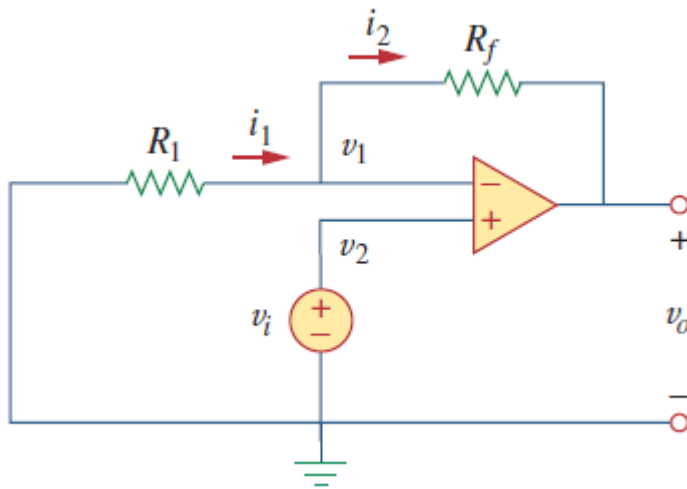
$$v_a - v_o = 12 - 2v_a \Rightarrow v_o = 3v_a - 12$$

But  $v_a = v_b = 2$  V for an ideal op amp, because of the zero voltage drop across the input terminals of the op amp. Hence,

$$v_o = 6 - 12 = -6 \text{ V}$$

# Noninverting Amplifier (I)

- Another important op amp circuit is the **noninverting amplifier** (同相放大器), which is designed to provide a **positive voltage gain**
- The basic configuration of the amplifier is the same as the inverting amplifier, except that the input and the ground are switched



- Once again applying KCL to the inverting terminal gives

$$i_1 = i_2 \Rightarrow \frac{0 - v_1}{R_1} = \frac{v_1 - v_o}{R_f}$$

$$\frac{-v_i}{R_1} = \frac{v_i - v_o}{R_f}$$

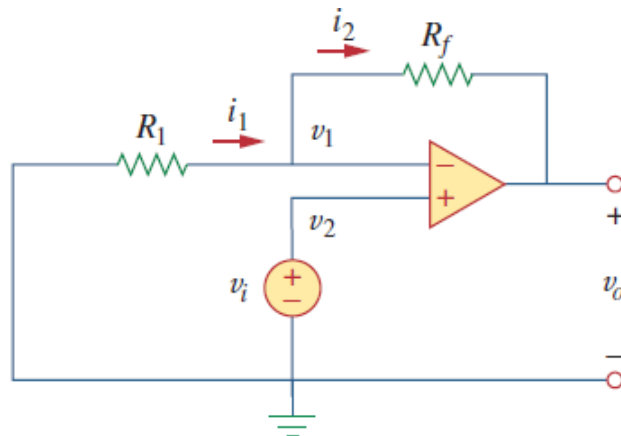


$$v_o = \left(1 + \frac{R_f}{R_1}\right)v_i$$

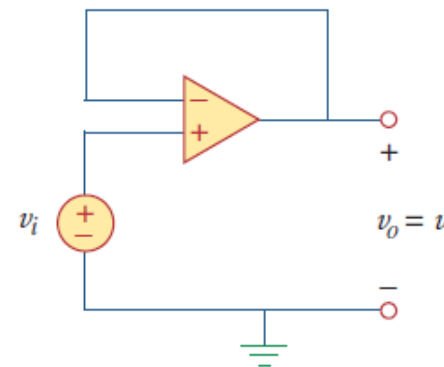


# Noninverting Amplifier (II)

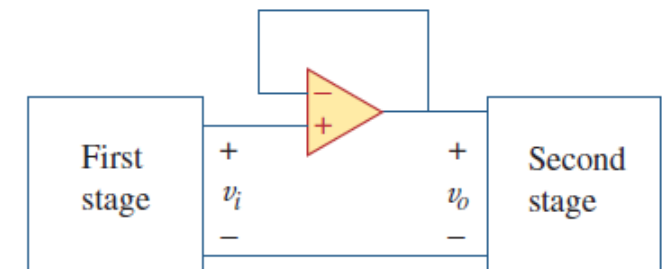
- The voltage gain is hence given by  $A_v = v_o/v_i = 1 + R_f/R_1$ 
  - Note that the gain here is positive, thus the amplifier is noninverting. Again, we notice that the gain depends only on the external resistors
- One aspect of this amplifier's gain is that it can never go below 1
  - One could replace the feedback resistor with a wire and disconnect the ground, i.e., make  $R_f = 0$  and  $R_1 = \infty$ , and the gain would still be 1
  - This configuration is called a **voltage follower** (电压跟随器) or a unity gain amplifier
  - It is good **for isolating** (隔离) two circuits while allowing a signal to pass through



**Figure 5.16**  
The noninverting amplifier.



**Figure 5.17**  
The voltage follower.



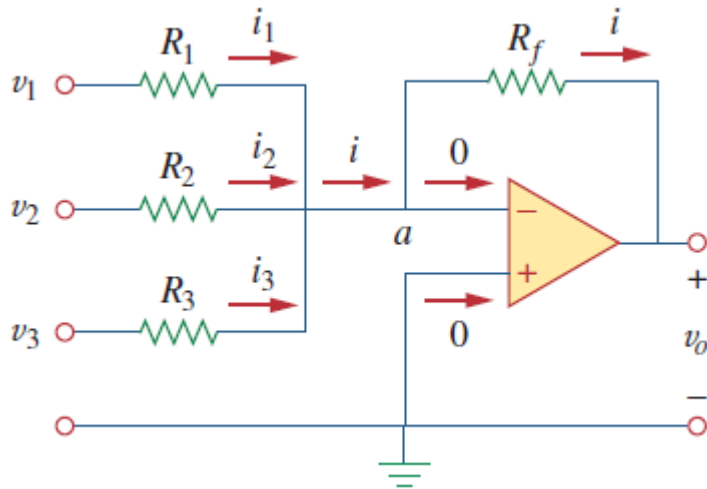
**Figure 5.18**  
A voltage follower used to isolate two cascaded stages of a circuit.

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- **Difference Amplifier**
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# Summing Amplifier

- Besides amplification, the op amp can perform addition by the summing amplifier and subtraction by the difference amplifier
- A **summing amplifier** (加法放大器) is an op amp circuit that combines several inputs and produces an output that is the weighted sum of the inputs
- The summing amplifier is a variation of the inverting amplifier. It takes advantage of the fact that the inverting configuration can handle many inputs at the same time



- If one takes the inverting amplifier and combines several inputs each via their own resistor, applying KCL at node  $a$  gives

$$i = i_1 + i_2 + i_3$$

$$i_1 = \frac{v_1 - v_a}{R_1}, \quad i_2 = \frac{v_2 - v_a}{R_2}$$

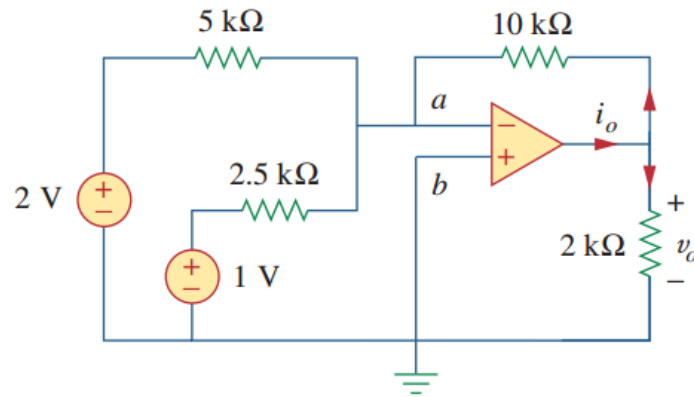
$$i_3 = \frac{v_3 - v_a}{R_3}, \quad i = \frac{v_a - v_o}{R_f}$$



$$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$$

# Example: Summing Amplifier

**Example 5.6** Calculate  $v_o$  and  $i_o$  in the op amp circuit in Fig. 5.22.



**Figure 5.22**  
For Example 5.6.

## Solution:

This is a summer with two inputs. Using Eq. (5.15) gives

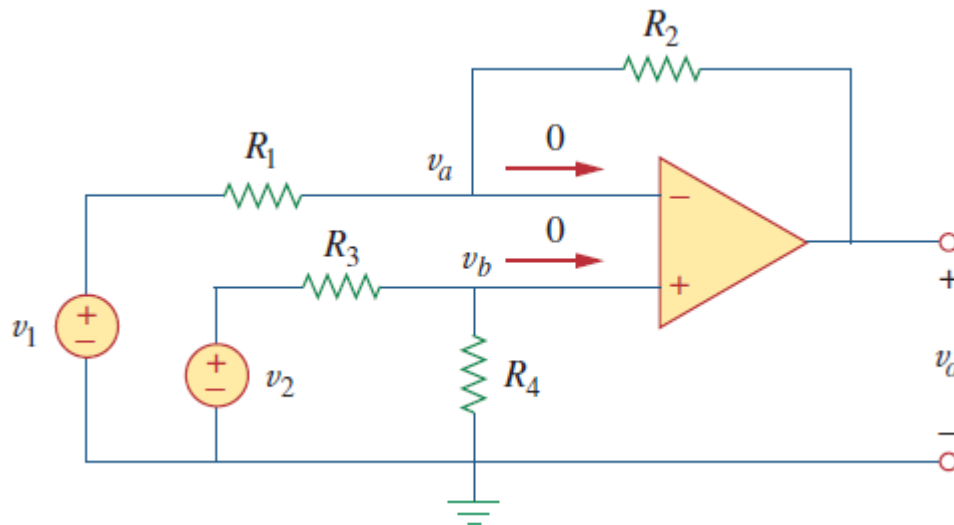
$$v_o = -\left[\frac{10}{5}(2) + \frac{10}{2.5}(1)\right] = -(4 + 4) = -8 \text{ V}$$

The current  $i_o$  is the sum of the currents through the 10-k $\Omega$  and 2-k $\Omega$  resistors. Both of these resistors have voltage  $v_o = -8 \text{ V}$  across them, since  $v_a = v_b = 0$ . Hence,

$$i_o = \frac{v_o - 0}{10} + \frac{v_o - 0}{2} \text{ mA} = -0.8 - 4 = -4.8 \text{ mA}$$

# Difference Amplifier (I)

- Subtraction should come naturally to the op amp since its output is proportional to the difference between the two inputs
- A **difference (or differential) amplifier (差分放大器)** is a device that amplifies the difference between two inputs but rejects any signals common to the two inputs



- Applying KCL to node a in the circuit

$$\frac{v_1 - v_a}{R_1} = \frac{v_a - v_o}{R_2}$$

$$\frac{v_2 - v_b}{R_3} = \frac{v_b - 0}{R_4}$$

$$v_o = \left( \frac{R_2}{R_1} + 1 \right) v_a - \frac{R_2}{R_1} v_1$$

$$v_b = \frac{R_4}{R_3 + R_4} v_2 \quad v_a = v_b$$

$$v_o = \left( \frac{R_2}{R_1} + 1 \right) \frac{R_4}{R_3 + R_4} v_2 - \frac{R_2}{R_1} v_1$$

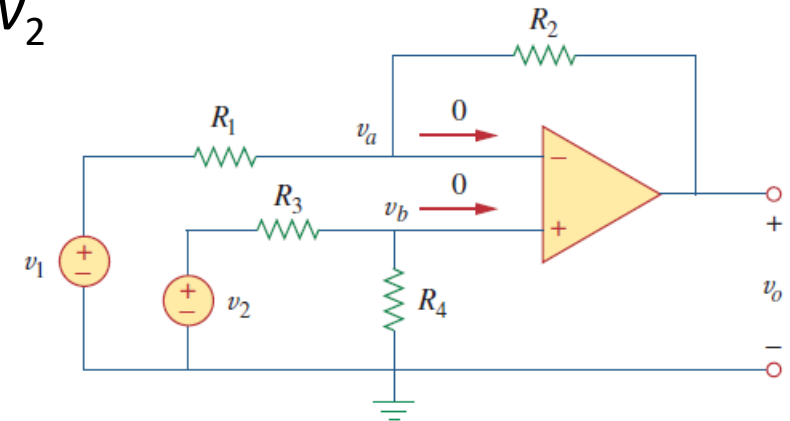
# Difference Amplifier (II)

- Since a difference amplifier must reject a signal common to the two inputs, the amplifier must have the property that  $v_o = 0$  when  $v_1 = v_2$
- For the given circuit, this is true if

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

- At which point, the output is

$$v_o = \frac{R_2}{R_1}(v_2 - v_1)$$



$$v_o = \left( \frac{R_2}{R_1} + 1 \right) \frac{R_4}{R_3 + R_4} v_2 - \frac{R_2}{R_1} v_1$$

- If  $R_1 = R_2$  and  $R_3 = R_4$ , the difference amplifier becomes a **subtractor (减法器)**, with the output

$$v_o = v_2 - v_1$$

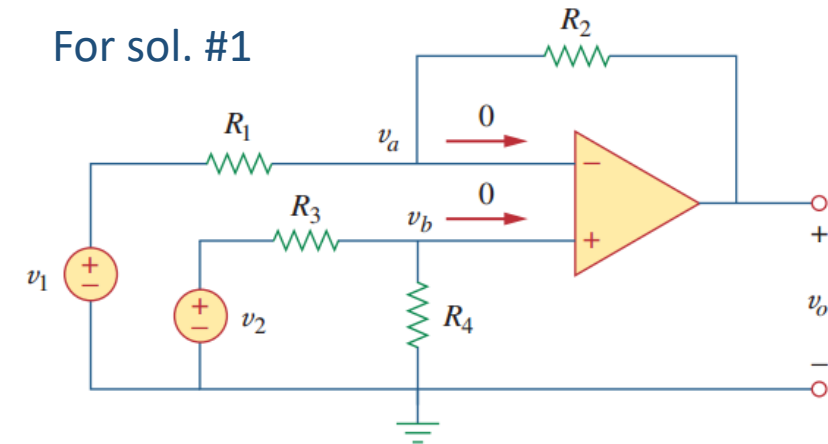
# Example: Difference Amplifier

## Example 5.7

**Design** an op amp circuit with inputs  $v_1$  and  $v_2$  such that  $v_o = -5v_1 + 3v_2$

### Solution #1:

Using only one op amp, as shown right, we can let  $R_2/R_1 = 5$  which means  $R_2 = 5R_1$ ; and  $R_2(1+R_1/R_2)/[R_1(1+R_3/R_4)] = 3$  which leads to  $R_3 = R_4$ . If we choose  $R_1 = 10 \text{ k}\Omega$  and  $R_3 = 20 \text{ k}\Omega$ , then  $R_2 = 50 \text{ k}\Omega$  and  $R_4 = 20 \text{ k}\Omega$ .



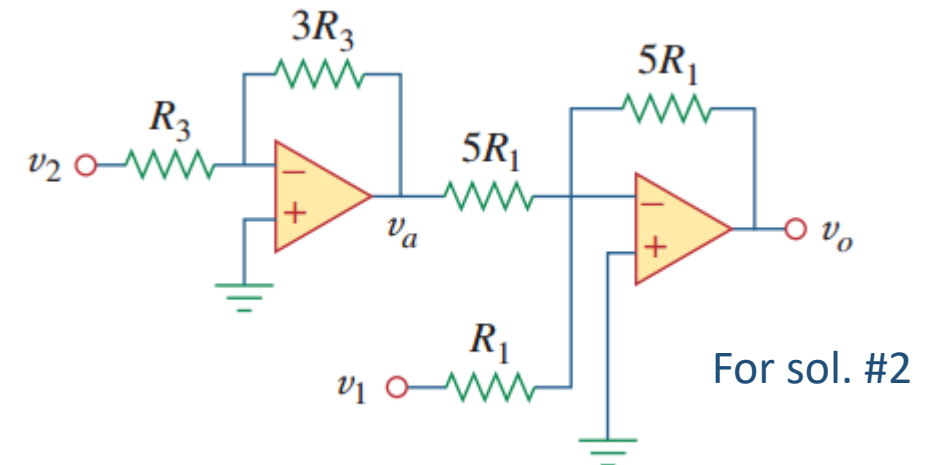
$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)}v_2 - \frac{R_2}{R_1}v_1$$

### Solution #2:

Using two op amps, we may cascade an inverting amplifier and a two-input inverting summer, as shown left.

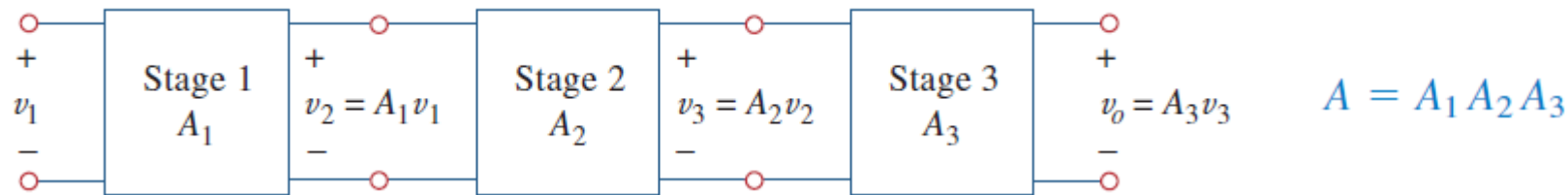
For the inverter  $v_a = -3v_2$

For the summer  $v_o = -v_a - 5v_1$



# Cascaded Op Amp Circuits

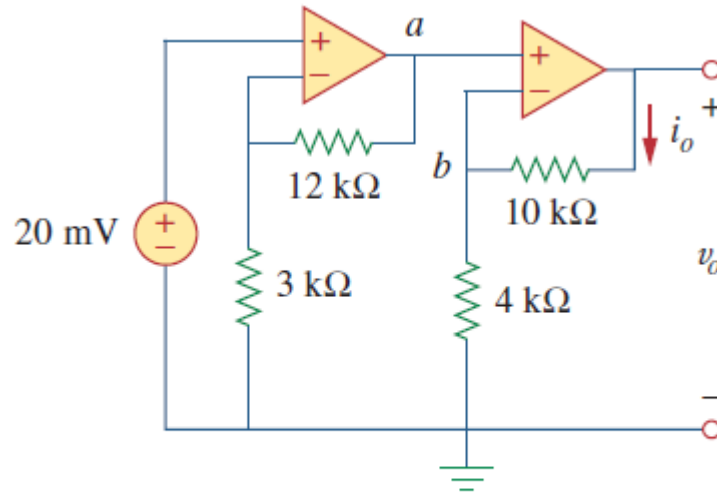
- Op amp circuits are modules or building blocks for designing complex circuits. It is often necessary in practical applications to connect op amp circuits in cascade (i.e., head to tail) to achieve a large **overall gain** (总增益)
- This head to tail configuration is called “**cascading**” (级联), each amplifier is then called a “**stage**”
- Due to the ideal op amps' infinite input and zero output resistance, stages can be chained together **without changing the input-output relationship of any one stage**
- Since the **output of one stage is the input to the next stage**, the overall gain of the cascade connection is the **product of the gains** of the individual op amp circuits





# Example1: Cascaded Op Amp Circuits

- Q: Find  $v_o$  and  $i_o$  in the circuit



**Figure 5.29**  
For Example 5.9.

Solution:

This is the **cascade of two noninverting amplifiers**.  
For the first,

$$v_a = \left(1 + \frac{12}{3}\right)(20) = 100 \text{ mV}$$

For the second,

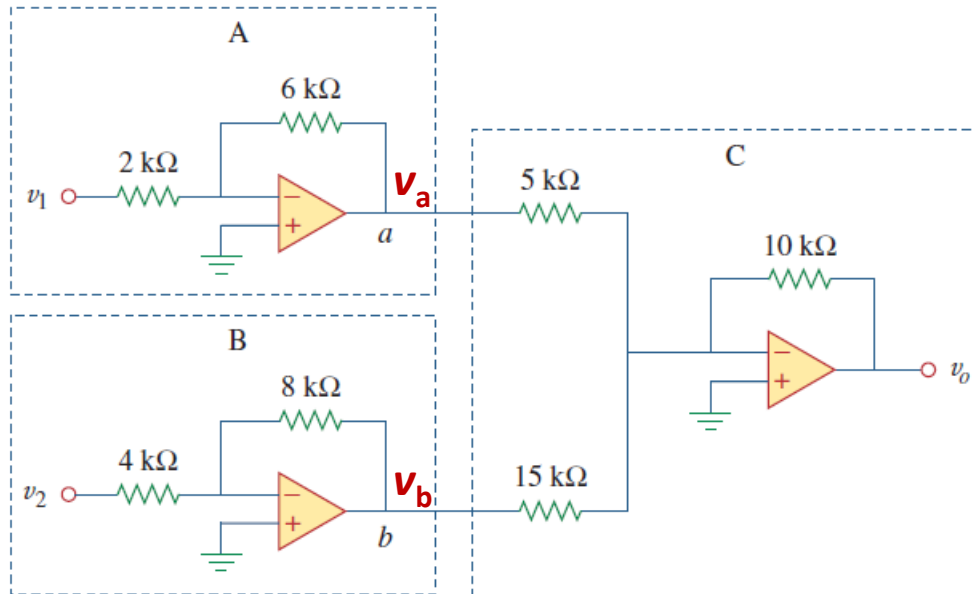
$$v_o = \left(1 + \frac{10}{4}\right)v_a = (1 + 2.5)100 = 350 \text{ mV}$$

Since  $v_b = v_a = 100 \text{ mV}$ ,

$$i_o = \frac{(350 - 100) \times 10^{-3}}{10 \times 10^3} = 25 \mu\text{A}$$

# Example2: Cascaded Op Amp Circuits

- Q: If  $v_1 = 1\text{ V}$  and  $v_2 = 2\text{ V}$ , find  $v_o$  in the op amp circuit



**Figure 5.31**  
For Example 5.10.

**Solution:**

The circuit composes of two stages and three sub-units, namely, two inverting and one summing amplifiers

For the inverting amplifier A

$$v_a = (-6\text{k}/2\text{k}) * v_1 = -3\text{ V}$$

For the inverting amplifier B

$$v_b = (-8\text{k}/4\text{k}) * v_2 = -4\text{ V}$$

For the summing amplifier C

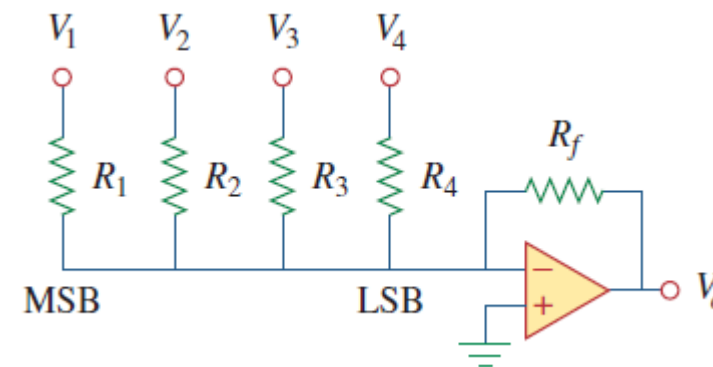
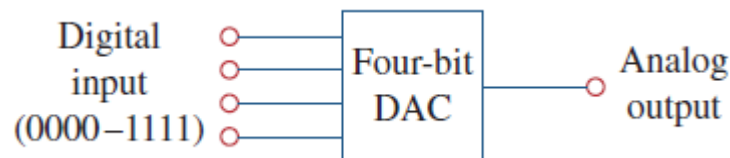
$$v_o = (-10\text{k}/5\text{k}) * v_a + (-10\text{k}/15\text{k}) * v_b = 6 + 2.667 = 8.667\text{ V}$$

# Outline

- Operational Amplifiers (Op Amps)
- Ideal Op Amp
- Inverting Amplifier
- Noninverting Amplifier
- Summing Amplifier
- Difference Amplifier
- Cascaded Op Amp Circuits
- **Applications**

# Applications: Digital-to-Analog Converter (I)

- The op amp is a fundamental building block in modern electronic instrumentation. It is used extensively in many devices, along with resistors and other elements
- The summing amplifier can be used to create a simple **digital-to-analog converter (DAC, 数模转换器)**, using the binary weighted ladder
  - In a digital signal, the input voltage will be either zero which represents '0' or a non-zero voltage which represents '1'
  - The function of a DAC is to take a series of binary values that represent a number and convert it to an analog voltage

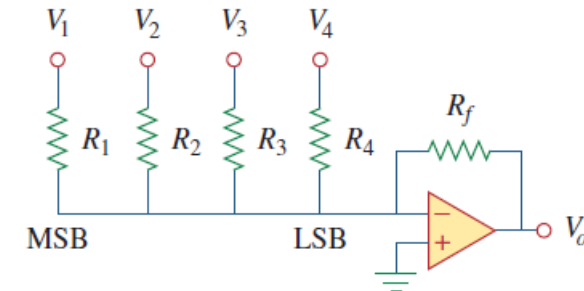


Binary weighted ladder type

# Applications: Digital-to-Analog Converter (II)

- By selecting the input resistors appropriately, each input will have a weighting according to the magnitude of their place value
  - Each lesser bit will have half the weight of the next higher bit
  - The feedback resistor provides an overall scaling, allowing the output to be adjusted according to the desired range
- According to the summing amplifier, the output can be related to the inputs by

$$-V_o = R_f \left( \frac{1}{R_1} V_1 + \frac{1}{R_2} V_2 + \frac{1}{R_3} V_3 + \frac{1}{R_4} V_4 \right)$$



- For example, if  $R_f = 10 \text{ k}\Omega$ ,  $R_1 = 10 \text{ k}\Omega$ ,  $R_2 = 20 \text{ k}\Omega$ ,  $R_3 = 40 \text{ k}\Omega$ ,  $R_4 = 80 \text{ k}\Omega$ , then we can relate the output to the inputs by

$$-V_o = V_1 + 0.5V_2 + 0.25V_3 + 0.125V_4$$

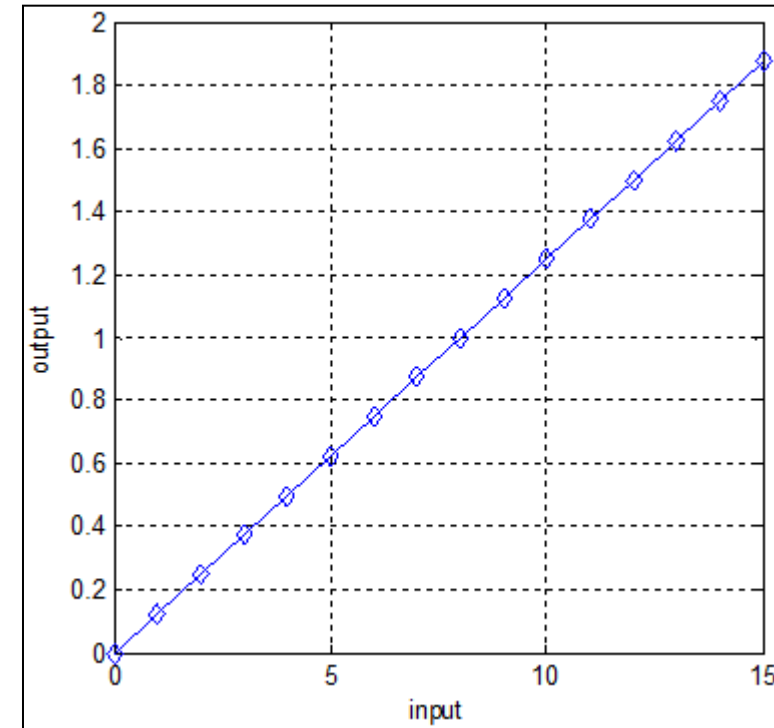
# Applications: Digital-to-Analog Converter (III)

- The table below shows the complete mapping between the inputs and output of the 4-bit DAC

**TABLE 5.2**

Input and output values of the four-bit DAC.

| Binary input<br>[ $V_1V_2V_3V_4$ ] | Decimal value | Output<br>$-V_o$ |
|------------------------------------|---------------|------------------|
| 0000                               | 0             | 0                |
| 0001                               | 1             | 0.125            |
| 0010                               | 2             | 0.25             |
| 0011                               | 3             | 0.375            |
| 0100                               | 4             | 0.5              |
| 0101                               | 5             | 0.625            |
| 0110                               | 6             | 0.75             |
| 0111                               | 7             | 0.875            |
| 1000                               | 8             | 1.0              |
| 1001                               | 9             | 1.125            |
| 1010                               | 10            | 1.25             |
| 1011                               | 11            | 1.375            |
| 1100                               | 12            | 1.5              |
| 1101                               | 13            | 1.625            |
| 1110                               | 14            | 1.75             |
| 1111                               | 15            | 1.875            |



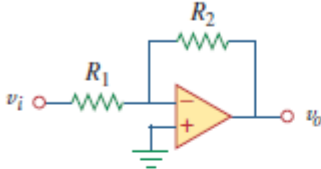
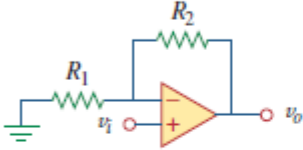
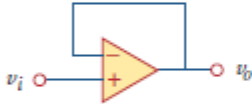
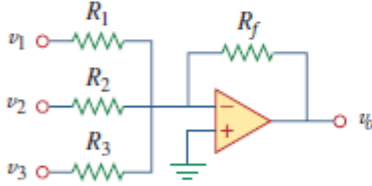
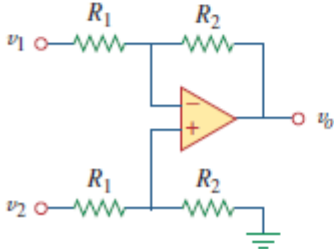
$$\begin{aligned}[V_1 V_2 V_3 V_4] &= [0010] &\Rightarrow &-V_o = 0.25 \text{ V} \\[V_1 V_2 V_3 V_4] &= [0011] &\Rightarrow &-V_o = 0.25 + 0.125 = 0.375 \text{ V} \\[V_1 V_2 V_3 V_4] &= [0100] &\Rightarrow &-V_o = 0.5 \text{ V} \\&\vdots && \\[V_1 V_2 V_3 V_4] &= [1111] &\Rightarrow &-V_o = 1 + 0.5 + 0.25 + 0.125 \\&&&= 1.875 \text{ V}\end{aligned}$$

# Summary

- The op amp is a high-gain amplifier that has high input resistance and low output resistance
- An ideal op amp has an infinite input resistance, a zero-output resistance, and an infinite gain
- For an ideal op amp, the current into each of its two input terminals is zero, and the voltage across its input terminals is negligibly small
- Table 5.3 summarizes the op amp circuits considered in this chapter. The expression for the gain of each amplifier circuit holds whether the inputs are dc, ac, or time-varying in general
- Op amp circuits may be cascaded without changing their input-output relationships
- Acknowledgement: This Lecture Note is partially based on materials from Prof. Chenchang Zan and partially adapted from the notes of the book “Fundamentals of Electric Circuits” by C. K. Alexander and M. N. O. Sadiku

**TABLE 5.3**

Summary of basic op amp circuits.

| Op amp circuit                                                                        | Name/output-input relationship                                                               |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|    | Inverting amplifier<br>$v_o = -\frac{R_2}{R_1}v_i$                                           |
|    | Noninverting amplifier<br>$v_o = \left(1 + \frac{R_2}{R_1}\right)v_i$                        |
|    | Voltage follower<br>$v_o = v_i$                                                              |
|   | Summer<br>$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$ |
|  | Difference amplifier<br>$v_o = \frac{R_2}{R_1}(v_2 - v_1)$                                   |